SARGUNAN C

EMBEDDED VALIDATION / FIRMWARE TEST ENGINEER

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PROFESSIONAL SUMMARY

B.Tech Electronics and Communication Engineering graduate (2026) targeting entry-level embedded firmware and validation roles. Built a C11 fault controller and a Python framework that validates it, both with public source code and live demos. Focused on boundary testing, debugging, and repeatable test automation.

TECHNICAL SKILLS

Programming: C11, Python

Firmware Development: modular C programming, finite-state machines, threshold handling, fault detection, fault recovery, error handling

Testing and Validation: unit testing, boundary-value testing, automated test execution, expected-versus-actual checks, regression testing, timeout and failure handling, JSON reporting

Embedded Fundamentals (conceptual; not hardware-tested): microcontroller architecture, registers, memory organization, GPIO, timers, interrupts, ADC, PWM

Communication Protocols (conceptual; not hardware-tested): UART, I2C, SPI

Tools: GCC, Git, GitHub, VS Code

SELECTED PROJECTS

Sensor Fault Controller

C11 · WebAssembly

GitHub Source | Live Demo

- Implemented a modular C11 controller with a NORMAL / WARNING / FAULT state machine, temperature thresholds, sensor-error detection, and latched over-temperature faults with recovery.
- Wrote a native C unit and boundary-test suite of 31 scenarios (31/31 passing) covering thresholds, sensor errors, latch/recovery sequences, and null-pointer handling.
- Compiled the same C source to freestanding WebAssembly for an interactive browser demo using simulated inputs; the same 31 scenarios pass (31/31).

Firmware Validation Framework

Python · C

GitHub Source | Live Dashboard

- Built a standard-library Python framework that runs the compiled C controller and checks exit code, state, fault code, and temperature against expected values.
- Wrote 37 controller test cases (37/37 passing) and 16 framework unit tests, including timeouts.
- Generated structured JSON reports and built an interactive dashboard to run regression suites.

INDUSTRIAL TRAINING

BSNL, Chennai — Industrial Training

Jun – Jul 2025

- Gained exposure to optical-fibre communication, broadband infrastructure, telecommunications equipment, network architecture, and network operations.

EDUCATION

B.Tech in Electronics and Communication Engineering

2022 – 2026

SASTRA Deemed University